CPRE/CYBE 2300: Cyber Security Fundamentals

Midterm: Thursday Oct. 8th, 2026 @ 7:45 AM

Student Name: Neal Sharma

Student ID Number: 406631026

Lab Section (Circle one): A (Tue 11:00) D (Tue 2:15) B (Thu 8:50) C (Thu 11:00)

1. [14 points; a) 10 & b) 4] Answer the following question:

a) **What are the network address and broadcast address of the following IP addresses**, and how are these addresses calculated using the given subnet mask?

i. 129.186.47.200/16 /16 ⇒ 2^{16}

Network: 129.186.0.0

Broadcast: 129.186.255.[redacted]254

ii. 192.168.0.77/27 /27 ⇒ 32 − 27 ⇒ 5 ⇒ 2^5 = 32 hosts

Network: 192.168.0.0

Broadcast: 192.168.0.[redacted]31

6

b) How many hosts can be attached to each of the above two networks?

i. Number of hosts is: 2^{16} = [Alot] ; I don't have a calculator

ii. Number of hosts is: $2^5 = 32 - 2$

c) One of the two addresses above cannot be used directly on the public Internet. Which one, and why?

The 192.188.0.77 can be used on the public internet of the 192 ip range is reserved for local usage only.

2

P.S. You do not need to compute the final numerical value. Simply show the calculation and leave it in its current form.

9

2. [15 points] Linux Permissions and Password Storage:

An administrator runs the command ls -l /etc/shadow and sees the following output:

- rw- r-- --- 1 root shadow 1123 Feb 2 16:13 /etc/shadow

The administrator then views a line inside the file: cpre2300:6h9ds8YCR$Rp9s/nYj$zsFg...

a. Based on the ls -l output, is this a file or a directory and why?

It is a file due to the 1st character (-). If it was a directory, the first character would have been 'd' ✓

b. Based on the ls -l output, who is allowed to read this file, and who is allowed to write to it?

The Owner is allowed to read and write
Group is allowed to read
Others have no permissions ✓

c. In the shadow file entry, what does the 6 between the first two dollar signs indicate 6?

It tells us the hashing algorithm used
In this case 6 is SHA-512 ✓

d. What is the string Kp3vR9xQ called, and how does it prevent an attacker who stole this file from using a standard, pre-computed Rainbow Table to find ta2300's password?

[This string is not in the hash that's given above]
But a salt is used to stop a rainbow table
∴ The string is a salt that is attached to the password ✓

e. Users cpre2300 and cybe2300 happen to choose exactly the same password. Will their hash fields in /etc/shadow be identical? Explain.

No since a salt is added to their passwords which makes the hashes different ✓

15

3. [18 points] Answer the following questions:

a) Identify three advantages of using Linux as a server compared to Windows.

It rarely crashes and is more reliable
It is faster and more efficient
It makes it easier to control every aspect of the server (windows is limited)
It is lightweight
Has a lot of packages which help with server management

b) ISU IT runs two services for remote students: (i) a live video stream of the CPR E 2300 lecture, and (ii) a download of the 3 GB Ubuntu Server installation image used in the labs. For each service, which transport-layer protocol would you choose, and why? What value would the Protocol field in the IP header contain for each of your choices?

(i) Live video:

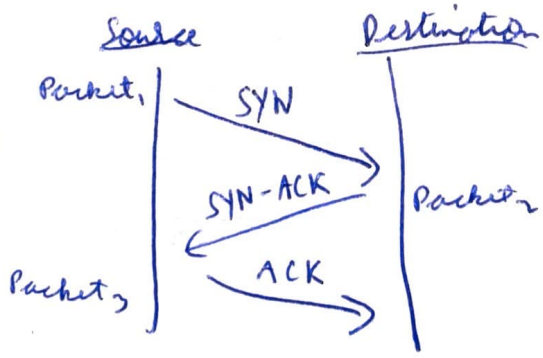
UDP → As its a live stream, the speed with which the student can see the stream matters more than reliability. UDP does not check if the student received the packet and is faster.
Value IP header will contain → 17

(ii) Image download:

TCP → TCP makes sure that no packet is lost or it confirms whether the student got the packet. If a packet gets lost, the image could get corrupted, which is why we don't use UDP.
Value → 6

c) Describe (or draw) the three-way handshake process and identify the protocol that uses it to establish a connection.

TCP uses a 3 way handshake



4. **[15 points] Linux Command Line Tools:**
Assuming access is limited to an Ubuntu Linux terminal with no graphical applications available, provide a single command (including necessary options) to complete the following tasks. Assume you are in your home directory.

a) Download a file from https://repo.anaconda.com/miniconda.sh and save it to your system specifically with the name my_script.sh

wget link my-script.sh

b) Open the file /var/log/messages using any text editor

nano /var/log/messages

c) Query the DNS for the IPv4 address of www.iastate.edu

dig www.iastate.edu

d) Send ICMP request packets to a computer with the IP address 1.2.3.1 to test connectivity.

ping 1.2.3.1

e) Display the path (every router hop) that packets take from your computer to 8.8.8.8

[man route and go from there (IDK)]

traceroute 8.8.8.8

-2

5. [12 points] Routing Table (Longest Prefix Match)

The following shows the output of the `route -n` command on Computer C. Answer the questions based only on this output (consider 0.0.0.0 under Gateway equivalent to a direct connection, and 0.0.0.0 under Destination as the default route).

Row	Destination	Gateway	Genmask	Flags	Iface
1	192.168.10.0	0.0.0.0 } direct	255.255.255.0	U	eth0
2	172.16.20.0	0.0.0.0 } direct	255.255.255.0	U	eth1
3	10.5.0.0	192.168.10.1	255.255.0.0	UG	eth0
4	10.5.12.0	172.16.20.1	255.255.255.0	UG	eth1
5	0.0.0.0 default	192.168.10.254	0.0.0.0	UG	eth0

If Computer C has an IP packet with one of the following destination addresses, explain the next device (Gateway IP) that C sends the packet to and the interface C uses. Refer to the row(s) of the table that match; if more than one row matches, state which row is used and why.

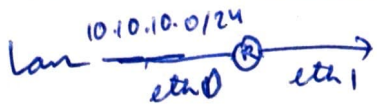
a. 192.168.10.77 → direct connection [0.0.0.0 gateway] eth0 interface ✓
as 192.168.10 is in row 1 with the direct gateway

b. 10.5.12.50 → This matches both rows 3 and 4 but row 4 is used [as its tighter]; ∴ 172.16.20.1 ; eth1 ✓

c. 10.5.120.9 → Row 3 is used ∴ 192.168.10.1 ; eth0 ✓

d. 172.16.2.15 → default route as 172.16.2 is not a destination ip in this route table. ∴ 192.168.10.254 ; eth0 ✓

12



6. [20 points] At Iowa State University (ISU), there is a firewall installed on a router with two interfaces: eth0 connects to the internal LAN (10.10.10.0/24) and eth1 connects to the external Internet. Provide the exact iptables commands to implement the following independent policies. Assume the table is flushed before each part and the default policy is ACCEPT.

Refer to the Reference Material if you want to referesh your memory with the Common iptables Syntax

a. Prevent all incoming SSH traffic originating from the Internet (eth1) from reaching the internal LAN.

iptables -A -FORWARD -p tcp -i eth1 -j DROP

pt 1. 4

b. Prevent the specific internal computer 10.10.10.50 from sending any UDP packets out to the Internet.

iptables -A - FORWARD -p udp -s 10.10.10.50 -o eth1 -j DROP

5

c. Prevent all internal computers (10.10.10.0/24) from accessing the external web server at IP 104.20.15.10 via HTTP.

iptables -A-FORWARD - s 10.10.10.0 -d 104.20.15.10 -dPort 80 -j DROP

iptables - A -FORWARD -s 104.20.15.10 -d 10.10.10.0 -dPort 80 - j DROP

5

d. Drop any packets originating from the Internet (eth1) that are trying to access the router's own internal services via SSH

~~iptables -A -INPUT -p tcp -dPort 22~~

iptables -A - INPUT -i eth1 -p tcp -dPort22 -j DROP

5

19

. **[9 points] Answer the following questions:**

a) Briefly explain the concept of Carrier Sense Multiple Access (CSMA)

CSMA is a connectivity protocol that sends data in a randomized way to different devices. If a collision is detected, it sends the packet again.

b) Traditional packet filtering firewalls make decisions based solely on individual packets (stateless), identify another type of packet filtering and briefly describe how it works.

Stateful firewalls. They keep the track of the state of current and previous connections. For eg: With TCP, a stateful firewall knows to allow the SYN-ACK after the initial SYN is sent.

c) A trojan on a victim's Linux laptop adds the following line to the file /etc/hosts:
66.66.66.93 login.bankofamerica.com
Explain what happens when the victim later browses to login.bankofamerica.com, and name this type of attack.

This is a DNS poisoning attack where an attacker makes a trustworthy DNS redirect to a malicious IP. They could simulate Bank of America to get user bank details.

Reference Material (Common iptables Syntax)

iptables [-t *table*] [-*operation chain*] [-p *protocol*] [-s *srcip*] [-d *dstip*] [-i *inif*] [-o *outif*] [–param1 value1 ...] -j *target*

- *table*: `filter`, `nat`, `mangle`
- *operation*: (first uppercase letter) Append, Delete, Insert, List, Flush, Policy, ...
- *chain*: `INPUT`, `OUPTUT`, `FORWARD`, `PREROUTING`, `POSTROUTING`
- *protocol*: `tcp`, `udp`, `icmp`, `all`, ...
- *srcip*, *dstip*: IP address, e.g. `1.1.1.1`, `2.2.2.0/24`
- *inif*, *outif*: interface name, e.g. `eth0`
- *param*, *value*: protocol specific parameter and value
  - `sport`, `dport`, `tcp-flags`, `icmp-type`, ...
- *target*: `ACCEPT`, `DROP`, `RETURN`, ...

Q	Earned	Total
Q 01	9	14
Q 02	15	15
Q 03	18	18
Q 04	13	15
Q 05	12	12
Q 06	19	20
Q 07	9	9
Total	95	103